BIOL-1 Practice Test 03

Module 7: Molecular Genetics

Instructions: This practice test covers material from Module 7. Answer all questions to the best of your ability.

Part A: Multiple Choice (25 questions)

Choose the best answer for each question.

1. The three parts of a nucleotide are:

A) Sugar, base, amino acid
B) Phosphate, sugar, nitrogenous base
C) Phosphate, lipid, protein
D) Base, ribose, peptide bond

2. In DNA, adenine (A) pairs with:

A) Guanine
B) Cytosine
C) Thymine
D) Uracil

3. How many hydrogen bonds form between guanine and cytosine?

A) 1
B) 2
C) 3
D) 4

4. What does "semi-conservative" mean in DNA replication?

A) Only half the DNA is copied

B) Each new molecule has one old strand and one new strand

C) Only one strand is used as a template

D) Replication happens at half speed

5. The enzyme that unwinds the DNA double helix is:

A) DNA polymerase

B) RNA polymerase

C) Helicase

D) Ligase

6. DNA polymerase builds new DNA in which direction?

A) $3' \rightarrow 5'$

B) $5' \rightarrow 3'$

C) Both directions equally

D) It varies by organism

7. Okazaki fragments are found on the:

A) Leading strand

B) Lagging strand

C) mRNA

D) tRNA

8. Which enzyme joins Okazaki fragments together?

A) Helicase

B) Primase

C) DNA polymerase

D) Ligase

9. Transcription produces:

A) DNA

B) mRNA

C) Protein

D) tRNA

10. Where does transcription occur in eukaryotic cells?

A) Cytoplasm

B) Ribosome

C) Nucleus

D) Cell membrane

11. In RNA, uracil (U) replaces which DNA base?

A) Adenine

B) Guanine

C) Cytosine

D) Thymine

12. If the DNA template strand reads 3′-TAC-5′, the mRNA codon would be:

A) TAC

B) AUG

C) ATG

D) UAC

13. The start codon is:

A) UAA

B) AUG

C) UGA

D) UAG

14. Which type of RNA carries amino acids to the ribosome?

A) mRNA

B) tRNA

C) rRNA

D) miRNA

15. A codon is made up of how many nucleotide bases?

A) 1

B) 2

C) 3

D) 4

16. Where does translation take place?

A) Nucleus

B) Ribosome

C) Golgi apparatus

D) Mitochondria

17. Which ribosome site is where the growing protein chain is held?

A) A site

B) P site

C) E site

D) S site

18. A mutation that changes one base to another is called a:

A) Frameshift mutation

B) Point mutation

C) Deletion

D) Insertion

19. Why are frameshift mutations usually more harmful than point mutations?

A) They only affect one codon

B) They shift every codon after the mutation

C) They don't change the protein

D) They are always silent

20. The Central Dogma of molecular biology is:

A) Protein → RNA → DNA
B) RNA → DNA → Protein
C) DNA → RNA → Protein
D) DNA → Protein → RNA

21. In eukaryotes, sections of mRNA that are removed before translation are called:

A) Exons
B) Introns
C) Codons
D) Anticodons

22. How many stop codons are there?

A) 1
B) 2
C) 3
D) 4

23. An anticodon is found on:

A) mRNA
B) DNA
C) tRNA
D) rRNA

24. The enzyme that builds mRNA during transcription is:

A) DNA polymerase
B) Helicase
C) RNA polymerase
D) Ligase

25. If a DNA strand reads 5′-TTAAGC-3′, the complementary strand is:

A) 5′-TTAAGC-3′
B) 3′-AATTCG-5′

C) 3′-UUAAGC-5′
D) 5′-AATTCG-3′

Part B: Fill in the Blank (10 questions)

26. The building block (monomer) of DNA is a _______.

27. DNA polymerase can only build new DNA in the _______ direction.

28. The process of copying DNA into mRNA is called _______.

29. The process of reading mRNA to build a protein is called _______.

30. The start codon _______ codes for the amino acid methionine.

31. The three stop codons are UAA, UAG, and _______.

32. A _______ mutation occurs when a base is inserted or deleted, shifting the reading frame.

33. In eukaryotes, _______ are the coding sections of mRNA that are kept after splicing.

34. The small RNA molecule that carries amino acids to the ribosome is _______.

35. Each new DNA molecule after replication has one old strand and one new strand; this is called _______ replication.

Part C: Short Answer (4 questions)

36. Write the mRNA sequence for this DNA template strand: 3′-TACGGGAAACT-5′. Then use a codon table to determine the amino acids.

37. Name three enzymes involved in DNA replication and describe what each one does.

38. Compare a point mutation and a frameshift mutation. Which is usually more harmful and why?

39. Explain the Central Dogma of molecular biology. Where does each step happen in a eukaryotic cell?

End of Practice Test 03